

MP3 TAG READER PROJECT

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Project Title: MP3 Tag Reader

Abstract

The MP3 Tag Reader project is developed in C language to read and edit metadata information present in MP3 audio files. The application extracts ID3 tags such as Title, Artist, Album, Year, Genre, and Comments from an MP3 file and displays them to the user. It also allows modification of selected tags without affecting the audio data.

Introduction

MP3 files contain metadata information known as ID3 tags. These tags store details about the audio file, such as song title, artist name, album, year, genre, and comments. The MP3 Tag Reader project provides a simple interface to read and edit these tags efficiently.

Objectives

- Read metadata information from MP3 files.
- Display ID3 tag details.
- Edit selected tag fields.
- Preserve original audio content.
- Provide user-friendly command-line operation.

Software Requirements

- Visual Studio Code
- GCC Compiler
- C Programming Language

Functionalities

View Tags

- Song Title
- Artist Name
- Album Name
- Year
- Comments

Edit Tags

- Title
- Artist
- Album
- Year
- Comments

Working Principle

1. Open the MP3 file.
2. Read the ID3v2 header.
3. Extract individual tag frames.
4. Display tag information.
5. Modify selected tags when requested.
6. Write updated tag information back to the MP3 file.

Features

- Reads MP3 metadata.
- Supports editing tag information.
- Simple command-line interface.
- Preserves audio data.
- Fast and efficient processing.

Advantages

- Easy metadata management.
- User-friendly operation.
- Lightweight implementation.
- No impact on audio quality.

Applications

- Music library management.
- Audio file organization.
- Metadata editing tools.

Result

The MP3 Tag Reader project successfully reads and edits ID3 metadata tags from MP3 files. The application displays required information and updates selected fields accurately.

Conclusion

The project demonstrates file handling, binary data processing, and metadata management using C programming. It provides an efficient solution for reading and editing MP3 tag information.